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Department of Computer Science and Engineering
B.Tech II - Semester-III
CS231 – Object Oriented Programming Practical Assignment
Lab Assignment - 13

1. Write a C++ program to read details of n students (roll no, name, 3 subject marks) and display them in a **well-formatted table** using **io manipulators** (setw, setprecision, left, right, fixed). Use getline() and get() functions of **iostream** for input handling.

2. Develop a **function template** analyze<T>() that takes an array of type T and computes: Mean, Median, Standard deviation. The program should:

1. Accept numeric data of **any one type**
2. Display results **formatted** using manipulator
3. Use **iostream functions** to control character-based input (like confirmation prompts).

3. Create a program that reads a customer's name, account number and balance amount. The print a mini bank statement formatted as:

- Balance displayed in fixed-point notation with 2 decimal places.
- Align data in neat columns
- Display a current symbol using manipulators.

Hint: Use **fixed()**, **setprecision()**, and **setw()**.

4. Input product details - name, quantity, and price per item. Display them in a bill/format using manipulators (**setw()**, **fixed()**, **setprecision()**) and functions of iostream classes like **width()**, **fill()** and **precision()**.

5. Template-Based Statistics Analyzer

Question:

Design a **C++ program using templates** to analyze an array of data values.

1. Define a **class template** `Statistics<T>` that stores an array of elements of type T.
 - Include member functions to calculate **sum**, **average**, and **maximum**.
2. Define a **function template** `compareValues<T>()` that returns the **greater** of two values.
3. In `main()`, demonstrate both templates for:
 - An `int` array (e.g., student scores)
 - A `double` array (e.g., product prices)
4. Display results with appropriate formatting using manipulators (`fixed`, `setprecision(2)`, etc.).

Sample Output:

Integer Data:

Sum = 280

Average = 70.00

Maximum = 92

Double Data:

Sum = 1023.50

Average = 255.88

Maximum = 499.75

Comparing 25 and 42 → Greater value: 42

6. Employee Payroll Slip Generator with Stream Flags

Write a C++ program that reads information for n employees (employee ID, name, hourly wage, and hours worked). Calculate each employee's gross pay and print a formatted payroll slip:

Use stream flags and manipulators (`showpos`, `internal`, `setfill`, `setw`, etc.) to display:

Names left-aligned, ID right-aligned in a field of width 8 padded with zeros

Pay values with a leading + sign and always 2 decimals

Display a warning in the output using `endl` and `flush` if gross pay falls below a threshold.

7. Template-Based Circular Buffer Visualizer

Develop a class template `CircularBuffer<T>` to store and manage a fixed-size buffer of any element type. Add member functions to insert, remove, and display the buffer contents. In the display function:

Print each element in a table, using custom fill characters and specified field width

After every 5 elements, insert a horizontal line using manipulators and custom padding/fill

Test with both `char` and `double` types, formatting double values to 2 precision and left-justifying char values.

8. Date and Time Table with User-Defined Manipulator

Create a user-defined manipulator called `formatDate` that, when applied to an output stream, prints the current date in the `YYYY-MM-DD` format (use only standard libraries). Build a program to read and store the dates (using strings) of `n` events and print all event names along with their dates using the custom manipulator for display. Use other manipulators (`setw`, `left`, `right`) for column alignment.